

Class Activity: 10.07.2026

Scenario 1: Student Academic Performance Dashboard

CODE:

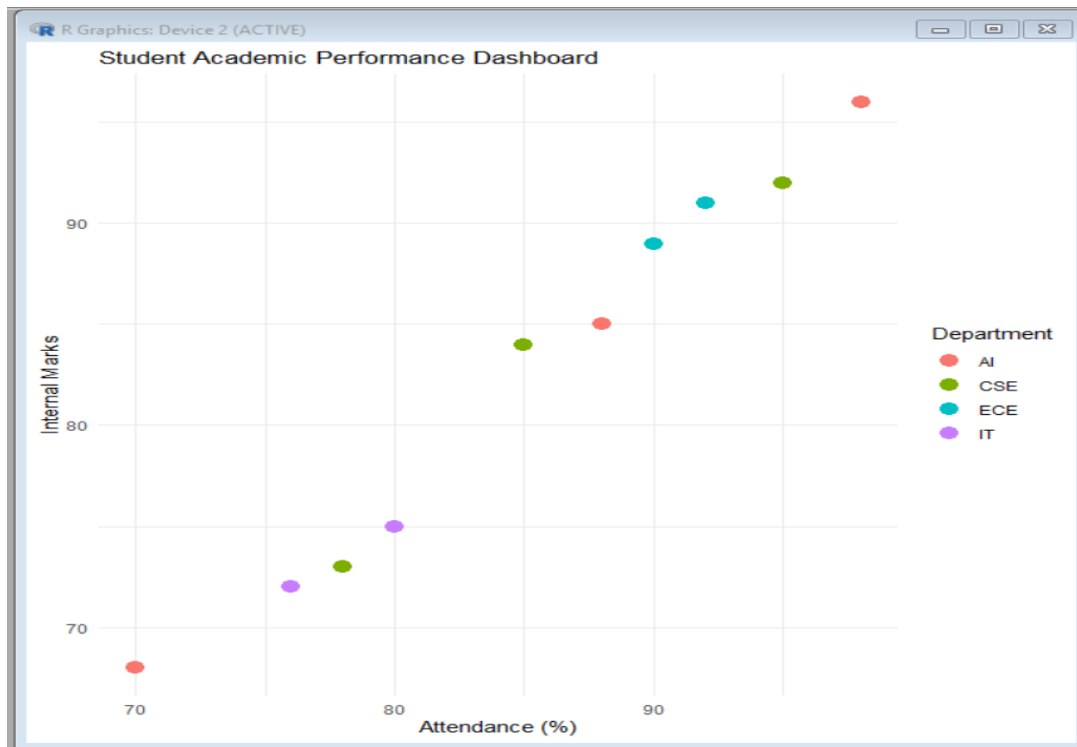
```
library(ggplot2)

student <- data.frame(

  StudentID = c(101,102,103,104,105,106,107,108,109,110),
  Attendance = c(95,88,76,90,85,70,80,92,78,98),
  Marks = c(92,85,72,89,84,68,75,91,73,96),
  Department = c("CSE","AI","IT","ECE","CSE","AI","IT","ECE","CSE","AI")
)

student

ggplot(student,
  aes(x = Attendance,
      y = Marks,
      color = Department)) +
  geom_point(size = 4) +
  labs(
    title = "Student Academic Performance Dashboard",
    x = "Attendance (%)",
    y = "Internal Marks",
    color = "Department"
  ) +
  theme_minimal()
```



Scenario 2: Electric Vehicle Market Analysis

CODE:

```
library(ggplot2)
```

```
ev <- data.frame(
```

```
  VehicleModel = c("Model A","Model B","Model C","Model D","Model  
E","Model F"),
```

```
  BatteryCapacity = c(35,45,55,65,75,85),
```

```
  Price = c(1200000,1500000,1800000,2200000,2600000,3000000),
```

```
  DrivingRange = c(250,300,350,400,450,500),
```

```
  Manufacturer = c("Tata","MG","Hyundai","Mahindra","BYD","Tata")
```

```
)
```

```
ev
```

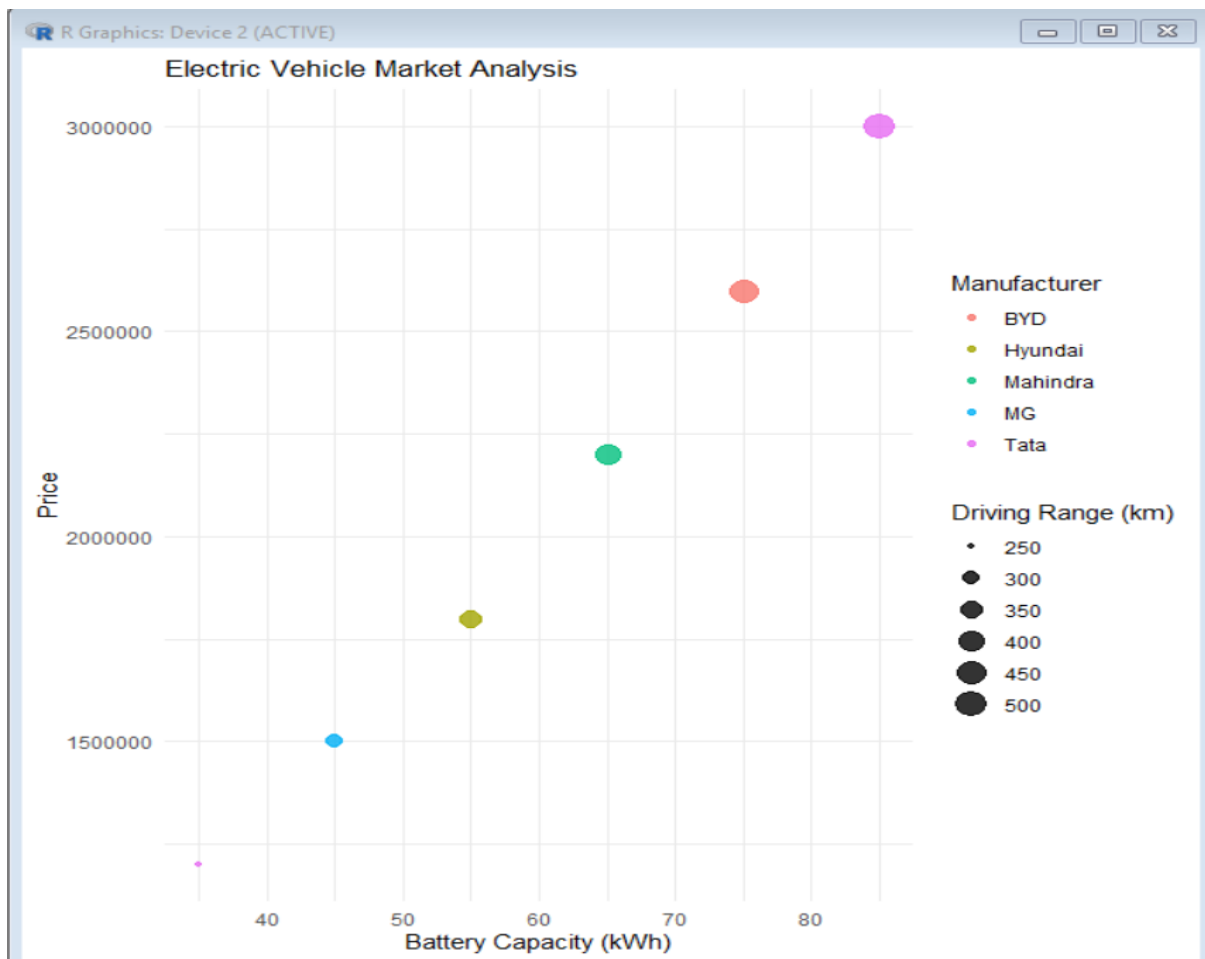
```
ggplot(ev,
```

```
  aes(x = BatteryCapacity,
```

```

y = Price,
size = DrivingRange,
color = Manufacturer)) +
geom_point(alpha = 0.8) +
labs(
  title = "Electric Vehicle Market Analysis",
  x = "Battery Capacity (kWh)",
  y = "Price",
  size = "Driving Range (km)",
  color = "Manufacturer"
) +
theme_minimal()

```



Scenario 3: Hospital Patient Admission Analysis

CODE:

```
library(ggplot2)

hospital <- data.frame(

  Department = c("Cardiology", "Neurology", "Orthopedics", "Pediatrics",
"General"),

  Patients = c(120, 90, 150, 100, 180),

  TreatmentCost = c(5000, 7000, 6000, 4000, 3500)

)

hospital

ggplot(hospital,

  aes(x = Department,

    y = Patients,

    fill = Department)) +

geom_bar(stat = "identity") +

geom_text(aes(label = TreatmentCost),

  vjust = -0.5,

  size = 4) +

labs(

  title = "Hospital Patient Admission Analysis",

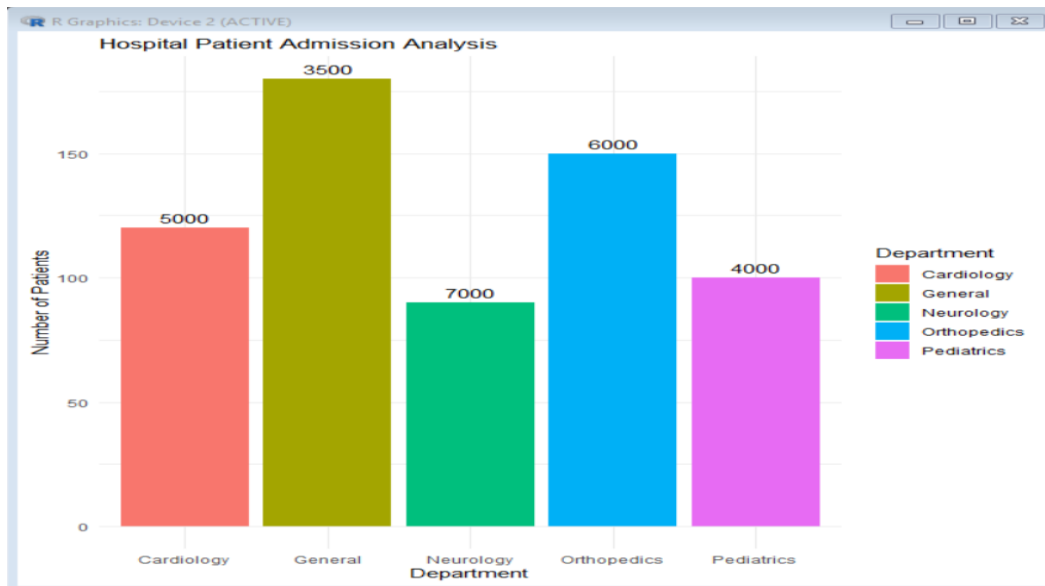
  x = "Department",

  y = "Number of Patients",

  fill = "Department"

) +

theme_minimal()
```



Scenario 4: Weather Monitoring System

CODE:

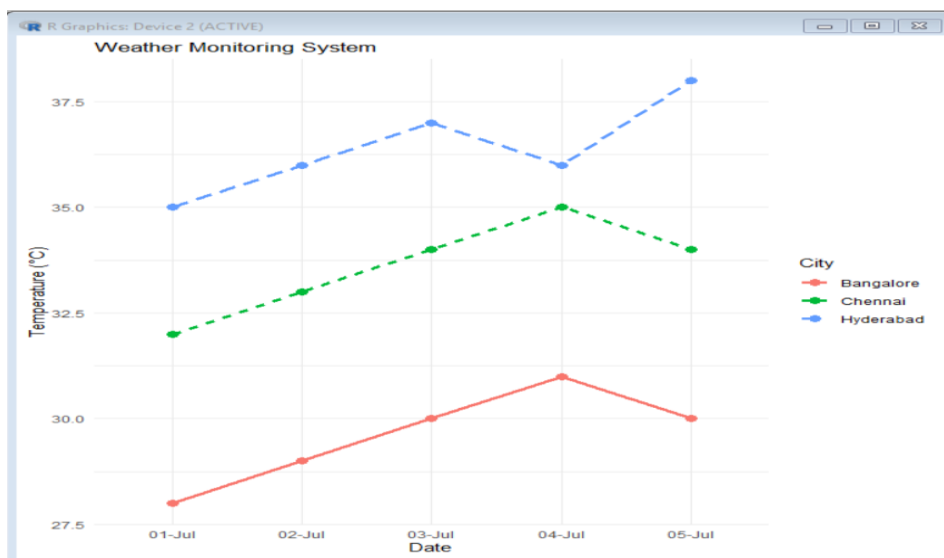
```
library(ggplot2)

weather <- data.frame(
  Date = c("01-Jul", "02-Jul", "03-Jul", "04-Jul", "05-Jul",
           "01-Jul", "02-Jul", "03-Jul", "04-Jul", "05-Jul",
           "01-Jul", "02-Jul", "03-Jul", "04-Jul", "05-Jul"),
  Temperature = c(32, 33, 34, 35, 34,
                  28, 29, 30, 31, 30,
                  35, 36, 37, 36, 38),
  City = c("Chennai", "Chennai", "Chennai", "Chennai", "Chennai",
           "Bangalore", "Bangalore", "Bangalore", "Bangalore", "Bangalore",
           "Hyderabad", "Hyderabad", "Hyderabad", "Hyderabad", "Hyderabad")
)

weather$Date <- factor(weather$Date,
                       levels = c("01-Jul", "02-Jul", "03-Jul", "04-Jul", "05-Jul"))
```

weather

```
ggplot(weather,  
  aes(x = Date,  
    y = Temperature,  
    color = City,  
    linetype = City,  
    group = City)) +  
  geom_line(size = 1.2) +  
  geom_point(size = 3) +  
  labs(  
    title = "Weather Monitoring System",  
    x = "Date",  
    y = "Temperature (°C)",  
    color = "City",  
    linetype = "City"  
  ) +  
  theme_minimal()
```



Scenario 5: Supermarket Sales Analysis

```
library(ggplot2)

sales <- data.frame(

  Month = c("Jan","Jan","Jan","Feb","Feb","Feb",
            "Mar","Mar","Mar","Apr","Apr","Apr"),
  ProductCategory = c("Groceries","Electronics","Clothing",
                      "Groceries","Electronics","Clothing",
                      "Groceries","Electronics","Clothing",
                      "Groceries","Electronics","Clothing"),
  SalesAmount = c(50000,30000,20000,
                  55000,32000,22000,
                  60000,35000,25000,
                  65000,38000,27000),
  Branch = c("Branch A","Branch B","Branch C",
             "Branch A","Branch B","Branch C",
             "Branch A","Branch B","Branch C",
             "Branch A","Branch B","Branch C")
)

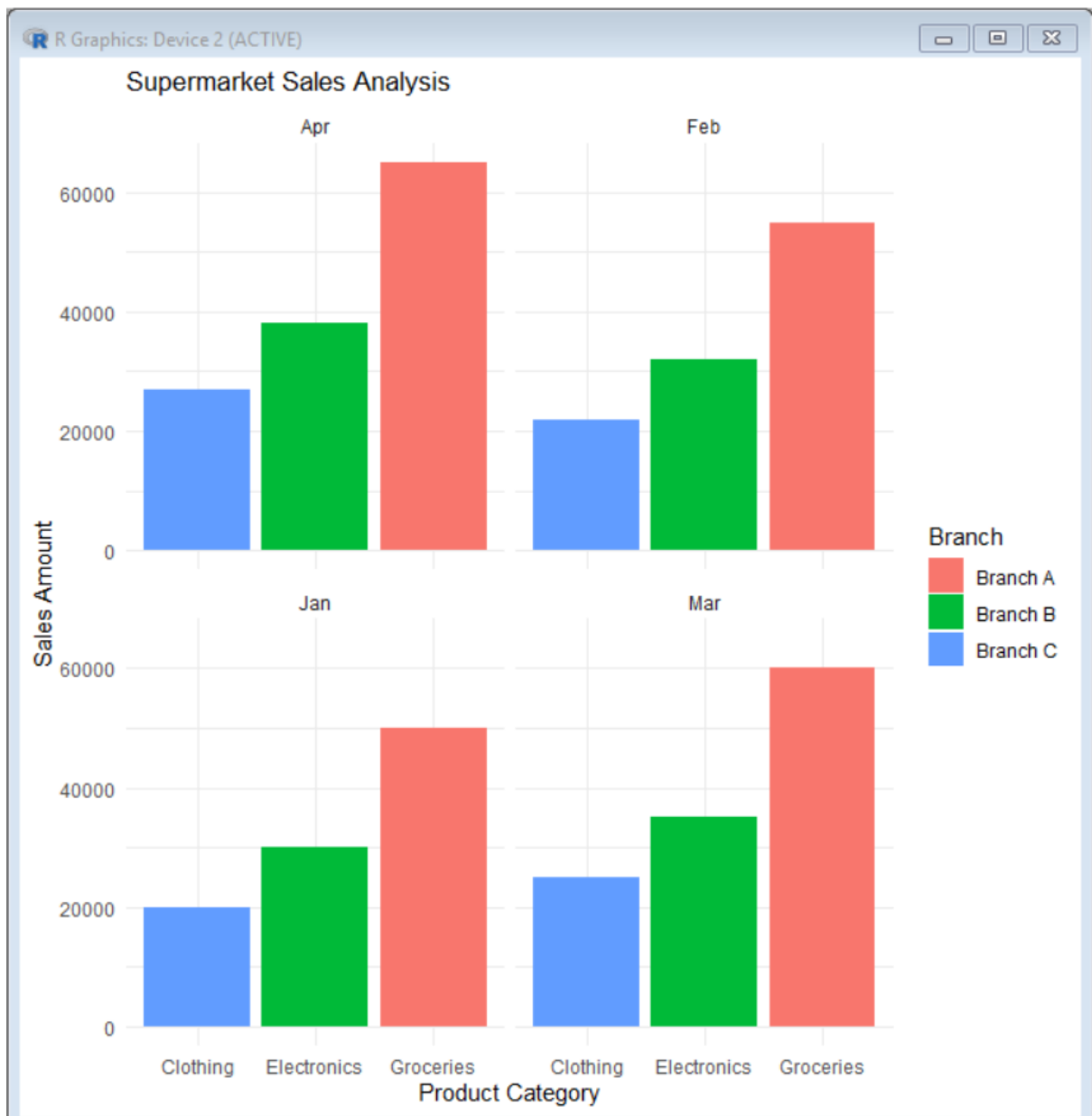
sales

ggplot(sales,
       aes(x = ProductCategory,
           y = SalesAmount,
           fill = Branch)) +
  geom_bar(stat = "identity",
           position = "dodge") +
```

```

facet_wrap(~Month) +
labs(
  title = "Supermarket Sales Analysis",
  x = "Product Category",
  y = "Sales Amount",
  fill = "Branch"
) +
theme_minimal()

```



Scenario 6: Employee Salary Distribution

CODE:

```
library(ggplot2)

employee <- data.frame(
  EmployeeID = c(101,102,103,104,105,106,107,108,109,110,
    111,112,113,114,115,116,117,118,119,120),
  Salary = c(25000,28000,30000,32000,35000,
    38000,40000,42000,45000,48000,
    50000,52000,55000,58000,60000,
    62000,65000,68000,70000,75000),
  Experience = c(1,2,2,3,4,5,5,6,7,8,
    8,9,10,10,11,12,13,14,15,16),
  Department = c("HR","Finance","IT","HR","Sales",
    "IT","Finance","Sales","HR","IT",
    "Finance","Sales","HR","IT","Finance",
    "Sales","HR","IT","Finance","Sales")
)

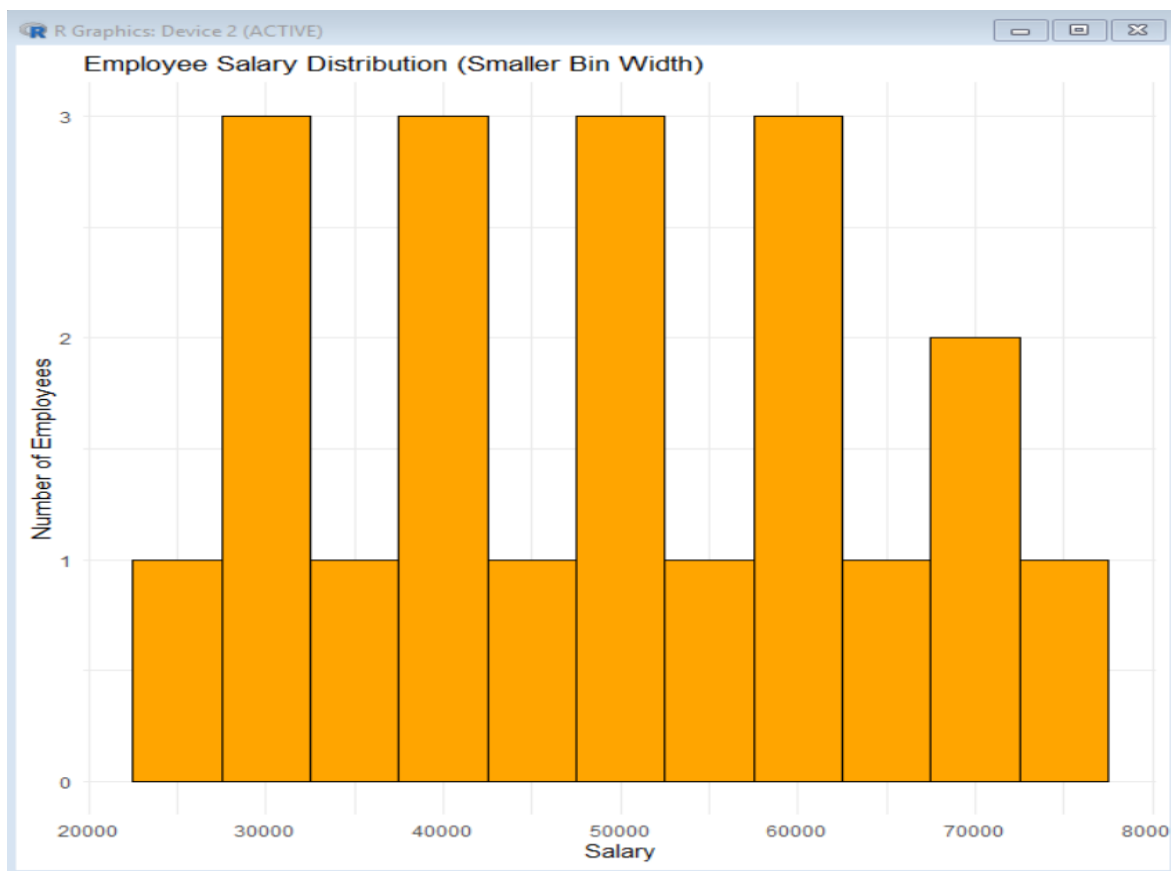
employee

ggplot(employee, aes(x = Salary)) +
  geom_histogram(binwidth = 10000,
    fill = "skyblue",
    color = "black") +
  labs(
    title = "Employee Salary Distribution",
    x = "Salary",
```

```

    y = "Number of Employees"
  ) +
  theme_minimal()
ggplot(employee, aes(x = Salary)) +
  geom_histogram(binwidth = 5000,
    fill = "orange",
    color = "black") +
  labs(
    title = "Employee Salary Distribution (Smaller Bin Width)",
    x = "Salary",
    y = "Number of Employees"
  ) +
  theme_minimal()

```



Scenario 7: Smartphone Market Comparison

CODE:

```
library(ggplot2)

smartphone <- data.frame(
  Brand = c("Samsung", "Apple", "OnePlus", "Xiaomi", "Realme",
            "Vivo", "Oppo", "Motorola"),
  RAM = c(8, 6, 12, 8, 6, 8, 12, 4),
  ProcessorSpeed = c(2.8, 3.2, 3.1, 2.9, 2.7, 2.8, 3.0, 2.5),
  Price = c(55000, 80000, 60000, 35000, 25000, 30000, 45000, 18000),
  Storage = c(128, 256, 256, 128, 128, 256, 512, 64)
)

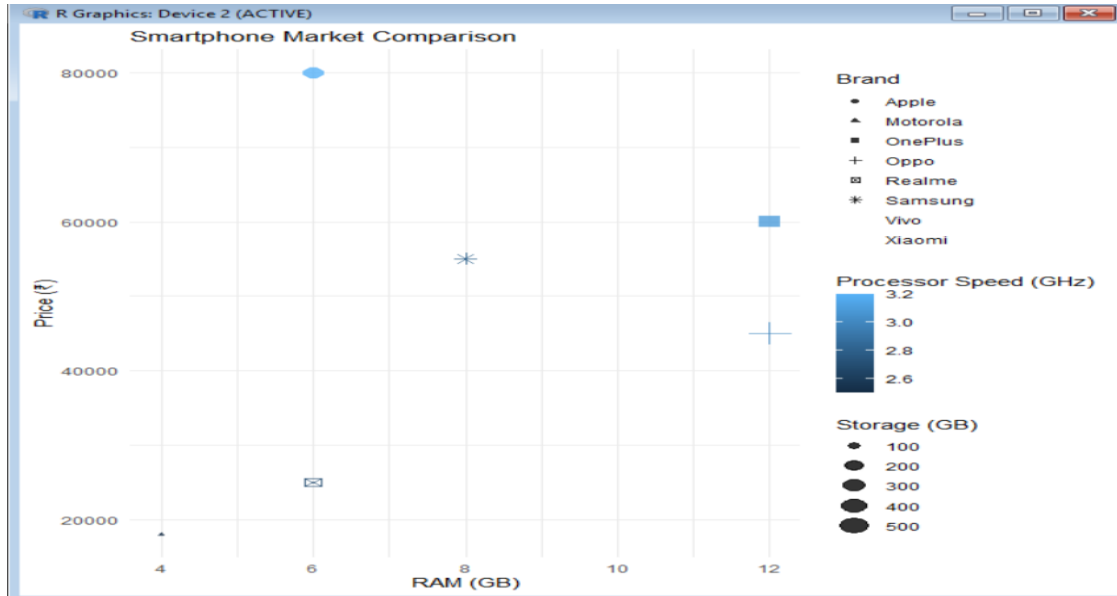
smartphone

ggplot(smartphone,
  aes(x = RAM,
      y = Price,
      shape = Brand,
      size = Storage,
      color = ProcessorSpeed)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Smartphone Market Comparison",
    x = "RAM (GB)",
    y = "Price (₹)",
    shape = "Brand",
    size = "Storage (GB)",
```

```

color = "Processor Speed (GHz)"
) +
theme_minimal()

```



Scenario 8: Online Movie Rating Analysis

CODE:

```

library(ggplot2)

movies <- data.frame(
  Genre = c("Action", "Comedy", "Drama", "Horror", "Romance"),
  AverageRating = c(4.5, 3.8, 4.2, 3.5, 4.0),
  NumberOfReviews = c(1200, 850, 950, 700, 1100),
  ReleaseYear = c(2021, 2022, 2023, 2021, 2022)
)

movies

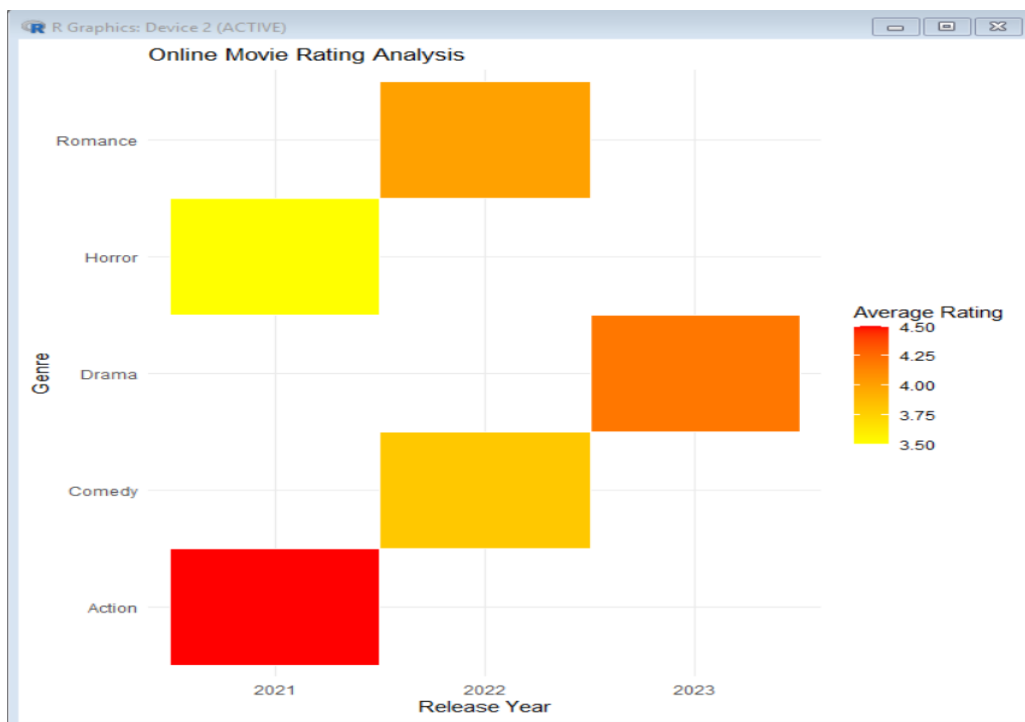
ggplot(movies,
  aes(x = factor(ReleaseYear),
    y = Genre,

```

```

    fill = AverageRating)) +
  geom_tile(color = "white") +
  scale_fill_gradient(
    low = "yellow",
    high = "red"
  ) +
  labs(
    title = "Online Movie Rating Analysis",
    x = "Release Year",
    y = "Genre",
    fill = "Average Rating"
  ) +
  theme_minimal()

```



Scenario 9: COVID-19 State-wise Analysis

CODE:

```
library(ggplot2)
```

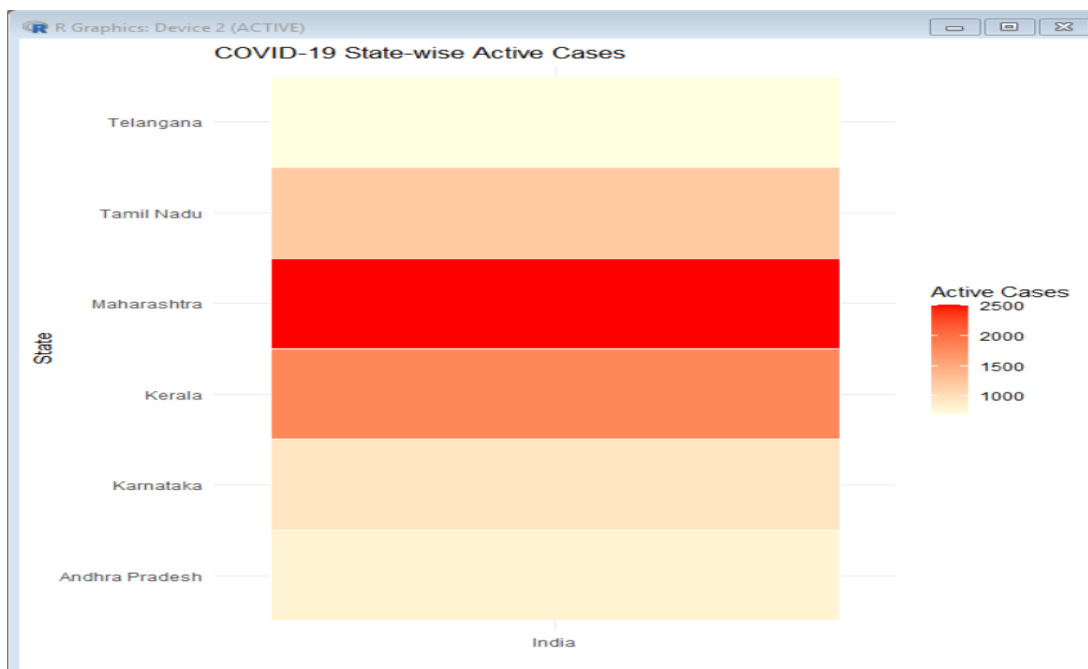
```
covid <- data.frame(
  State = c("Tamil Nadu",
            "Karnataka",
            "Kerala",
            "Maharashtra",
            "Andhra Pradesh",
            "Telangana"),
  ActiveCases = c(1200,950,1800,2500,800,700),
  VaccinationPercentage = c(92,90,94,91,89,88),
  Population = c(72000000,
                68000000,
                35000000,
                124000000,
                53000000,
                39000000)
)
```

```
covid
ggplot(covid,
  aes(x = "India",
      y = State,
      fill = ActiveCases)) +
  geom_tile(color = "white") +
  scale_fill_gradient(
    low = "lightyellow",
    high = "red"
```

```

) +
labs(
  title = "COVID-19 State-wise Active Cases",
  x = "",
  y = "State",
  fill = "Active Cases"
) +
theme_minimal()

```



Scenario 10: Iris Flower Classification

CODE:

```

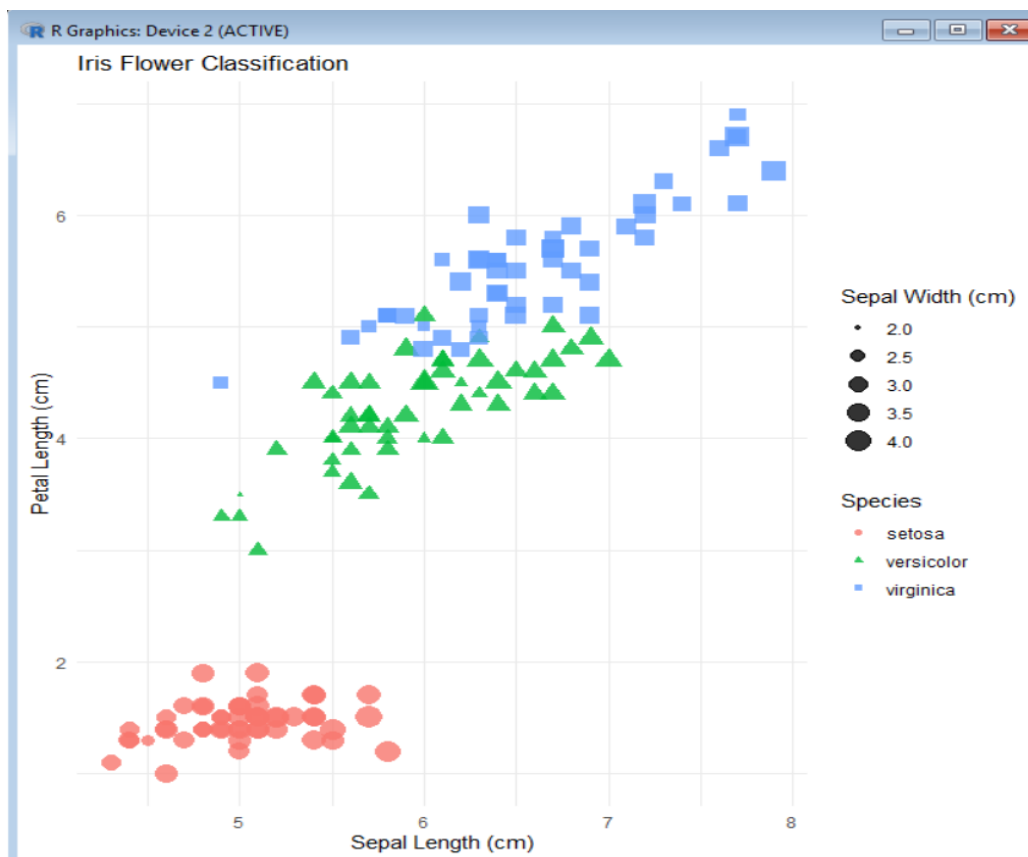
library(ggplot2)
data(iris)
head(iris)
ggplot(iris,
  aes(x = Sepal.Length,
    y = Petal.Length,

```

```

    color = Species,
    shape = Species,
    size = Sepal.Width)) +
geom_point(alpha = 0.8) +
labs(
  title = "Iris Flower Classification",
  x = "Sepal Length (cm)",
  y = "Petal Length (cm)",
  color = "Species",
  shape = "Species",
  size = "Sepal Width (cm)"
) +
theme_minimal()

```



Scenario Continuous Variables		Discrete Variables
1	Attendance, Marks	Department
2	Battery Capacity, Price, Driving Range	Manufacturer
3	Patients, Treatment Cost	Department
4	Temperature	Date, City
5	Sales Amount	Month, Product Category, Branch
6	Salary, Experience	Department
7	RAM, Price, Processor Speed, Storage	Brand
8	Average Rating, Number of Reviews	Genre, Release Year
9	Active Cases, Vaccination Percentage, Population	State
10	Sepal Length, Sepal Width, Petal Length, Petal Width	Species